Lesson 4.6 — Multiplying and Dividing Rational Expressions

Big idea: Multiply by multiplying tops and bottoms, then simplify. To divide, flip the second fraction and multiply.

Quick review

Multiply: (a/b) · (c/d) = (ac)/(bd). Best practice: factor first, cancel anything that matches across the multiplication, then multiply.
Divide: (a/b) ÷ (c/d) = (a/b) · (d/c). Flip the second, then multiply as above.
Domain: exclude every x that makes any denominator zero — including the divisor before flipping.

Worked example

Worked example. Multiply $[(x^2 – 4)/(x + 1)] \cdot [(x^2 – 1)/(x – 2)]$.

Factor: $[(x – 2)(x + 2)/(x + 1)] \cdot [(x – 1)(x + 1)/(x – 2)]$. Cancel (x – 2) and (x + 1): (x + 2)(x – 1).

Warm-ups

1. Compute $(3/x) \cdot (x/6)$.
2. Compute $(x^2/2) \div (x/4)$.
3. Compute $(2/x) \div (4/x^2)$.

Practice

1. Multiply $[(x – 3)/(x + 2)] \cdot [(x^2 – 4)/(x^2 – 9)]$.
2. Divide $[(x^2 – x – 6)/(x – 1)] \div [(x – 3)/(x^2 – 1)]$.
3. Multiply $[(x + 1)/(x^2 – 25)] \cdot [(x – 5)/(x + 1)]$.

Challenge

1. Simplify $[(x^2 – 9)/(x^2 + 6x + 9)] \cdot [(x^2 + 4x + 3)/(x – 3)]$.
2. Simplify $[(2x – 4)/(x^2 – x – 12)] \div [(x^2 – 4)/(x – 4)]$.

Answer key — Lesson 4.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 1/2.
2. $(x^2/2) \cdot (4/x) = 2x$.
3. $(2/x) \cdot (x^2/4) = x/2$.

Practice

1. $[(x - 3)/(x + 2)] \cdot [(x - 2)(x + 2)/((x - 3)(x + 3))] = (x - 2)/(x + 3)$.
2. $[(x - 3)(x + 2)/(x - 1)] \cdot [(x - 1)(x + 1)/(x - 3)] = (x + 2)(x + 1)$.
3. $[(x + 1)/((x - 5)(x + 5))] \cdot [(x - 5)/(x + 1)] = 1/(x + 5)$.

Challenge

1. $[(x - 3)(x + 3)/(x + 3)^2] \cdot [(x + 3)(x + 1)/(x - 3)] = (x + 1)$.
2. $[2(x - 2)/((x - 4)(x + 3))] \cdot [(x - 4)/((x - 2)(x + 2))] = 2/[(x + 3)(x + 2)]$.